

Speech Processing: BSAI - 7th Semester

Lab 5 Task Report (QUIZ)

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Objective

To implement and analyze the Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) in MATLAB, and to study the differences in their computation and performance.

Task 1: Calculation of DFT

MATLAB Code:

```
1 clc
2 clear all;
3 close all;
4 a = round(144/2);
5 N = length(a);
6
7 %Calculate DFT
8 for k=1:N
9     y(k)=0;
10    for n=1:N
11        y(k) = y(k)+a(n)*exp((-1i*2*pi*(k-1))*(n-1)/N);
12    end
13 end
14
15 k = 1:N
16 disp('The result is: ');y
17 figure(1);
18 subplot(2,1,1);
19 stem(k,abs(y(k)));
20 ylabel('Amplitude');
21 xlabel('Sample value of n');
22 title('Magnitude response of the DFT');
23 subplot(2,1,2);
24 stem(k,angle(y(k))*180/pi);
25 xlabel('Sample values n');
26 ylabel('Phase');
27 title('Phase response of the DFT');
```

Output Figure:

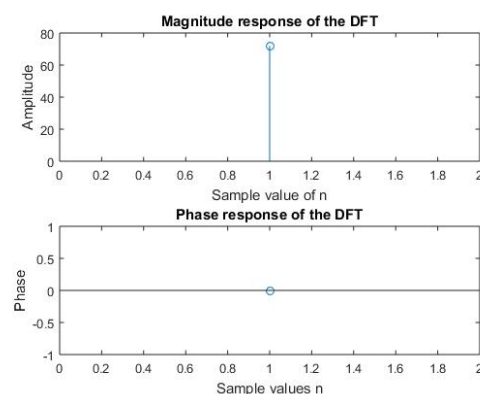


Figure 1: Magnitude and Phase Response of DFT

Explanation: This program computes the Discrete Fourier Transform (DFT) of the given sequence using its mathematical definition. Here, the signal is converted from time domain to frequency domain. The upper subplot represents the magnitude spectrum of the DFT, and the lower subplot shows its phase response in degrees.

Task 2: Calculation of FFT

MATLAB Code:

```

1 %Calculate FFT
2
3 clc;
4 clear all;
5 close all;
6 t = -2:1:2;
7 y=[zeros(1,2) 1 zeros(1,2)];
8 subplot (3,1,1);
9 stem(t,y);
10 grid;
11 title('Impulse Response');
12 xlabel('Time');
13 ylabel('Amplitude');
14 N = round(144/2);
15 xk = fft(y,N);
16 magxk = abs(xk);
17 angxk = angle(xk);
18 k = 0:N-1
19 subplot(3,1,2);
20 stem(k,magxk);
21 xlabel('k');
22 ylabel('|x(k)');
23 subplot(3,1,3);
24 stem(k,angxk);
25 xlabel('k');
26 ylabel('arg(x(k))');

```

Output Figure:

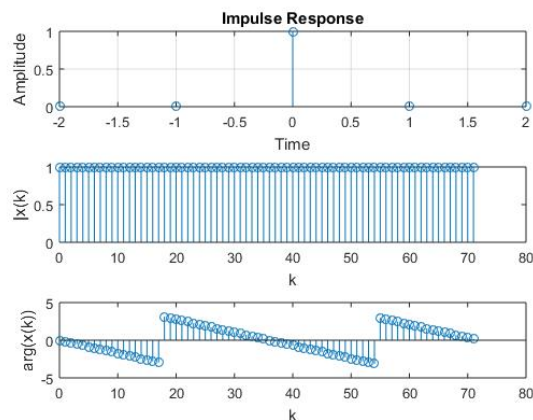


Figure 2: Fast Fourier Transform (FFT) of Impulse Signal

Explanation: In this code, the FFT of an impulse signal is computed. The first subplot shows the discrete impulse signal in time domain, while the next two subplots display the FFT's magnitude and phase spectra. The FFT provides a computationally efficient method for obtaining the same result as DFT but with reduced execution time.

Conclusion

In this lab, both DFT and FFT were implemented in MATLAB. The DFT provided a direct mathematical understanding of the frequency transformation, while the FFT demonstrated the efficiency of fast algorithms in frequency-domain computation. Through this experiment, the concepts of spectral analysis and frequency representation of signals were reinforced.